



RESEARCH ARTICLE

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David A. Carozza and Mathieu Boudreault contributed equally to this work.

A Global Flood Risk Modeling Framework Built With Climate Models and Machine Learning

David A. Carozza¹  and Mathieu Boudreault¹ 

¹Département de Mathématiques, Université du Québec à Montréal, Montréal, QC, Canada

Key Points:

- We present a global flood model built using machine learning methods fitted with historical flood occurrences and impacts
- Forced with a climate model, the global flood model is fast, flexible and consistent with global climate
- We provide global flood hazard (occurrence) and risk (population displaced) maps over 4,734 watersheds

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

D. A. Carozza,
david.carozza@gmail.com

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Author Contributions:

Conceptualization: David A. Carozza, Mathieu Boudreault

Data curation: David A. Carozza

Formal analysis: David A. Carozza, Mathieu Boudreault

Funding acquisition: David A. Carozza, Mathieu Boudreault

Investigation: David A. Carozza, Mathieu Boudreault

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Abstract Large scale flood risk analyses are fundamental to many applications requiring national or international overviews of flood risk. While large-scale climate patterns such as teleconnections and climate change become important at this scale, it remains a challenge to represent the local hydrological cycle over various watersheds in a manner that is physically consistent with climate. As a result, global models tend to suffer from a lack of available scenarios and flexibility that are key for planners, relief organizations, regulators, and the financial services industry to analyze the socioeconomic, demographic, and climatic factors affecting exposure. Here we introduce a data-driven, global, fast, flexible, and climate-consistent flood risk modeling framework for applications that do not necessarily require high-resolution flood mapping. We use statistical and machine learning methods to examine the relationship between historical flood occurrence and impact from the Dartmouth Flood Observatory (1985–2017), and climatic, watershed, and socioeconomic factors for 4,734 HydroSHEDS watersheds globally. Using bias-corrected output from the NCAR CESM Large Ensemble (1980–2020), and the fitted statistical relationships, we simulate 1 million years of events worldwide along with the population displaced in each event. We discuss potential applications of the model and present global flood hazard and risk maps. The main value of this global flood model lies in its ability to quickly simulate realistic flood events at a resolution that is useful for large-scale socioeconomic and financial planning, yet we expect it to be useful to climate and natural hazard scientists who are interested in socioeconomic impacts of climate.

Plain Language Summary Flood is among the deadliest and most damaging natural disasters. To protect against large scale flood risk, stakeholders need to understand how floods can occur and their potential impacts. Stakeholders rely on global flood models to provide them with plausible flood scenarios around the world. For a flood model to operate at the global scale, climate effects must be represented in addition to hydrological ones to demonstrate how rivers can overflow throughout the world each year. Global flood models often lack the flexibility and variety of scenarios required by many stakeholders because they are computationally demanding. Designed for applications where detailed local flood impacts are not required, we introduce a rapid and flexible global flood model that can generate hundreds of thousands of scenarios everywhere in the world in a matter of minutes. The model is based on a historical flood database from 1985 to 2017 that is represented using an algorithm that learns from the data. With this model, the output from a global climate model is used to simulate a large sample of floods for risk analyses that are coherent with global climate. Maps of the annual average number of floods and number of displaced people illustrate the models results.

1. Introduction

Flood is consistently among the most damaging natural disasters in terms of economic losses (Gall et al., 2009) and mortality (Hu et al., 2018). Impacts generated by flood result from a complex set of interactions between climatic, hydrological, demographic, and economic factors. Despite improvements in flood defenses and other technologies resulting in reduced vulnerability (Paprotny et al., 2018), nominal flood-related economic losses have increased rapidly in recent decades due to developments in exposure such as total wealth and urban area (Jongman et al., 2012), and rising prices. After normalizing relative to exposure, Barredo (2009) and Neumayer and Barthel (2011) did not identify statistically significant increasing trends in economic flood losses, yet short time series, challenges with data, and the inability to control for changes in flood defenses challenged these studies. Trends in insured losses, which are further complicated by the extent to which exposure is insured, were not found for atmospheric natural disasters at the global scale

Methodology: David A. Carozza, Mathieu Boudreault
Project Administration: Mathieu Boudreault
Resources: Mathieu Boudreault
Software: David A. Carozza
Supervision: Mathieu Boudreault
Validation: David A. Carozza, Mathieu Boudreault
Visualization: David A. Carozza, Mathieu Boudreault
Writing – original draft: David A. Carozza, Mathieu Boudreault
Writing – review & editing: David A. Carozza, Mathieu Boudreault

(Barthel & Neumayer, 2011). However, the same study detected a positive trend in normalized flood-driven insured losses from 1980 to 2007 in the United States. Paprotny et al. (2018) reconstructed floods in Europe since 1870 and uncovered increasing trends in normalized flooded area and persons affected but decreases in normalized fatalities and financial losses in recent decades.

Large scale flood risk analyses are pivotal to disaster management and relief planning at regional, national, and international levels. Flood risk analyses can build resilience by informing investment needs in mitigation and financial mechanisms such as insurance (Vorogushyn et al., 2018). By the nature of their business, insurance and reinsurance companies are heavily exposed to flood risk across the globe through private and/or public insurance programs (OECD, 2016), whereas banks are subject to mortgage defaults following floods (FRBSF, 2019; Ouazad & Kahn, 2019). With mounting pressure coming from regulators and other bodies worldwide, the financial services industry (banks, insurers, and reinsurers) will soon need to disclose and stress test their solvency and stability to various climate scenarios (Bank of England, 2019; Task Force on Climate-related Financial Disclosures, 2017), which includes how future flood risk will affect their profitability.

A major methodological challenge to designing global models is to guarantee that flood risk is consistent from climate, hydrological, hydraulic, and exposure standpoints, such that physically-consistent global climate patterns drive the local hydrological cycle over many watersheds (Vorogushyn et al., 2018). This is a particularly important issue for the financial services industry as their global portfolios are impacted by large-scale climate patterns affecting people over distances of thousands of kilometers. Top down approaches typically force low-resolution hydrological models with meteorological or climate model outputs that simulate runoff that is consistent with simulated climate patterns (Winsemius et al., 2013; Yamazaki et al., 2011). Such approaches are not just global because they represent the entire planet, but because connections between watersheds in space that are driven by climate are resolved. Top down methods have been used to delve into large-scale flood risk questions such as examining patterns of interannual climate variability (Ward et al., 2014) and to project the impacts of future climate and socioeconomic change (Dottori et al., 2018; Jongman et al., 2014; Ward et al., 2020). An important weakness of this approach is its lack of focus on flooding occurrence and impact in itself, and the inability to resolve small scale floods.

Bottom up approaches consider higher resolution processes that employ a combination of rainfall-runoff or hydrological modeling to drive a hydraulic component and calculate flood damage over watersheds (de Bruijn et al., 2014; Falter et al., 2016; Sampson et al., 2015). These models are typically forced by meteorological (historical, simulated, or projected) or discharge distributions. This more detailed approach is closer to assessing localized impacts of flood but is challenged by high computational demands and data requirements that are not necessarily available globally (Ward et al., 2015). For both approaches, the number of scenarios available is limited and they lack the flexibility required by planners, relief organizations, regulators, and the financial services industry to analyze the socioeconomic, demographic, and climatic factors affecting exposure.

In this paper, we introduce a data-driven, global, fast, flexible, and climate-consistent flood risk modeling framework for applications that do not necessarily require high-resolution flood mapping. Our framework is unique in that it is driven by historical flood and environmental observations. It takes advantage of the speed of statistical models to quickly generate large global catalogs of flood events that are physically consistent with climate. Distributions of occurrence and impact can then be analyzed in terms of climatic and socioeconomic factors and over spatial scales of interest. The framework is therefore capable of examining interannual climate variability and looking into the future, accounting for global change over various greenhouse gas emission and socioeconomic scenarios, in addition to accounting for climate-driven connections between watersheds. The model will be of interest to flood researchers for socioeconomic impact studies, climatic research of the impacts on population or wealth affected, risk analyses in poorly sampled watersheds (Hrachowitz et al., 2013), and stress testing risk portfolios for the financial services industry.

We overview the model in Section 2. Sections 3 and 4 present the statistical model framework and the simulation of a global catalog, respectively. We evaluate the quality and realism of the flood model in Section 5, present global flood hazard and risk maps in Section 6, and conclude in Section 7. Supporting information is available online that presents additional validations and descriptions.

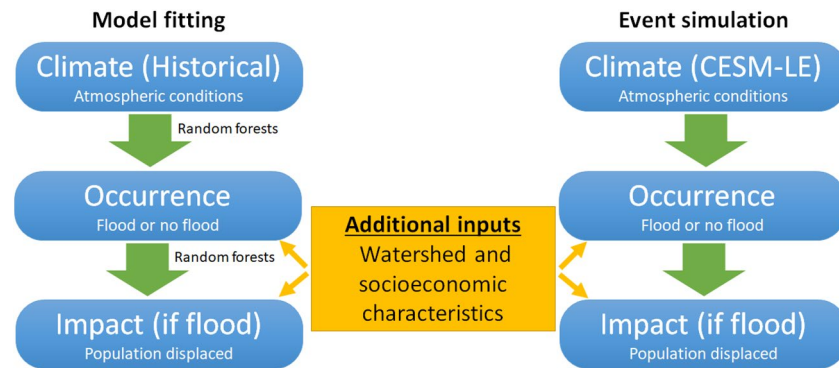


Figure 1. Global flood model schematic. The fitting process is detailed on the left, whereas the simulation process is presented on the right.

2. Model at a Glance

The global flood-risk modeling framework presented here consists of two key stages that are summarized in Figure 1: (1) statistical model estimation, and (2) event simulation. As a first step, we built flood occurrence and impact databases by intersecting a spatial database of historical flood events, the Dartmouth Flood Observatory Global Active Archive of Large Flood Events (DFO) (Brakenridge, 2010), with the HydroBASINS (Lehner & Grill, 2013) dataset of global watersheds at Pfafstetter level 5. This resulted in a historical event set of watershed-floods, that represent floods over multiple individual watersheds. In addition to a spatial representation of flood events, the DFO dataset provides a number of impact measures. We used the population displaced to quantify flood impact in terms of the fraction of the watershed population displaced.

We then constructed novel databases used for statistical modeling by associating the occurrence and intensity of each watershed-flood event to the driving environmental and socioeconomic conditions of the watershed affected. To capture multiple potential drivers of flood, we considered historical precipitation and temperature as climatic predictors, watershed information such as topography, land use, and soil type, as well as socioeconomic characteristics of the watershed such as population density and GDP per capita. Table 1 details the predictor variables used to construct our historical flood database. We explored both classical and machine/statistical learning methods to fit the occurrence and impact models, and proceeded to use the random forest as our overall favored model.

To generate a large global catalog of synthetic flood events, we then stochastically simulated from the occurrence and impact models. Our approach is flexible enough to support any observational or modeled temperature and precipitation data; however, to expand upon the limited observational record (Munoz & Dee, 2017) we generate a large sample of flood occurrence probabilities and impacts using bias-corrected precipitation and temperature output from the National Center for Atmospheric Research's (NCAR) Community Earth System Model (CESM) Large Ensemble (LE) (Kay et al., 2015). For each of the 40 ensemble members, the climate model output (Table 2) provides us with simulated temperature and precipitation for the historical period from 1980 to 2020, and so provided 1,600 years of physically consistent climate variables at the global scale. Sampling from the statistical models using outputs from the CESM-LE thus form the basis of the global model output; 1 million years of events worldwide.

3. Flood Model Estimation

Riverine floods are generally driven by an excess of precipitation less evapotranspiration relative to the storage capacity of the watershed. As such, they are typically caused by intense and widespread precipitation that saturates a watershed's soils and drives water to overflow a river's banks. Such floods can similarly be caused by heavy snowmelt. Although we do not treat the case of ice jam floods, riverine floods can be caused by the accumulation of ice blocking the flow of the river. Riverine flood impacts are then determined by the interaction of overflowed water with population and property exposed.

Table 1
Response and Predictor Variables Used in Statistical Fits of Occurrence and Impact Models

Database	Variable	Data reference
Response variables		
DFO - Dartmouth Flood Observatory Global Active Archive of Large Flood Events		Brakenridge (2010)
	Flood event (4,499 polygons)	
	Population displaced (4,499 polygons with 3,160 observations)	
Predictor variables		
CHIRPS - Climate Hazards Group Infrared Precipitation with Stations		Funk (2015)
	Precipitation (Daily from 1981-present, 0.05°) For sites between $[-50^{\circ}, +50^{\circ}]$ latitude	
CPC Precipitation - CPC Global Unified Gauge-Based Analysis of Daily Precipitation		Xie et al. (2007)
	Precipitation (Daily from 1979-present, 0.5°) For sites outside $[-50^{\circ}, +50^{\circ}]$ latitude	
CPC Temperature - CPC Global Daily Temperature		Shi (2007)
	Temperature (Daily from 1979-present, 0.5°)	
HydroBASINS Watersheds		Lehner and Grill (2013)
	Watershed length (4,734 watershed polygons)	
	Watershed width (4,734 watershed polygons)	
	Watershed area (4,734 watershed polygons)	
	Watershed upstream area (4,734 watershed polygons)	
	Latitude of watershed centroid (4,734 watershed polygons)	
	Longitude of watershed centroid (4,734 watershed polygons)	
HydroBASINS Digital Elevation Model		Lehner and Grill (2013)
	Aspect (30 arcsec)	
	Elevation (30 arcsec)	
	Hillslope (30 arcsec)	
	Slope (30 arcsec)	
	Topographic position index (30 arcsec)	
TI - High-resolution global topographic index values		Marthews et al. (2015a)
	Topographic index (15 arcsec)	
GLC-SHARE - Global Land Cover SHARE		Latham et al. (2014)
	Artificial surfaces (30 arcsec)	
	Cropland (30 arcsec)	
	Grassland (30 arcsec)	
	Tree covered areas (30 arcsec)	
	Shrubs covered areas (30 arcsec)	
	Herbaceous vegetation, aquatic, or regularly flooded (30 arcsec)	
	Mangroves (30 arcsec)	
	Sparse vegetation (30 arcsec)	
	Bare soil (30 arcsec)	
	Snow and glaciers (30 arcsec)	
	Water bodies (30 arcsec)	
GSD - Global Soil Dataset for Earth System Modeling		Shangguan et al. (2014)

Table 1
Continued

Database	Variable	Data reference
GDB - Global depth to bedrock	Sand (30 arcsec)	Shangguan et al. (2017)
	Silt (30 arcsec)	
	Clay (30 arcsec)	
	Gravel (30 arcsec)	
	Bulk soil density (30 arcsec)	
GLHYMPS 2.0 - GLobal HYdrogeology MaPS 2.0	Global depth to bedrock (30 arcsec)	Gleeson et al. (2014)
	Bedrock porosity (1,394,912 polygons)	
	Bedrock permeability (1,394,912 polygons)	
GPW - Gridded Population of the World	Population count (Annual, 2.5 arcmin)	CIESIN (2018)
	Population count (Annual, 5 arcmin) For temporal extrapolation of GPW to 1980	
GDPHDI - Gridded global datasets for GDP and HDI over 1990–2015	Gross domestic product (PPP) per capita (Annual, 5 arcmin)	Kummu et al. (2019)

Note. Dataset temporal and spatial resolution are in parentheses.

We have built a statistical framework for flood risk that drives flood occurrence and impact with environmental and socioeconomic predictor variables (Table 1) at the watershed scale. Wolock et al. (2004) and Rumsey et al. (2015) used a similar statistical framework to define hydrologic-landscape regions and estimate base-flow, respectively. In a study that also applied the DFO (Brakenridge, 2010) database, Li et al. (2019) used machine-learning models on a record of historical flood occurrence to construct global flood susceptibility maps. To the authors' knowledge, such an empirical approach has not been applied to model flood risk at the global scale. Our approach falls into the class of lumped hydrological models (Bevin, 2012; Perrin et al., 2013) since the forcing data we used is averaged over a watershed. However, instead of considering a system output quantity such as discharge, we directly model watershed (1) flood hazard and (2) impact. We achieve this by building databases and statistical models of (1) flood occurrence and (2) the fraction of population displaced, and express them in terms of environmental and demographic predictor variables.

3.1. Observational Data

Observations for 4,499 historical flood events (Figure 2) were available in the Dartmouth Flood Observatory (DFO) Global Active Archive of Large Flood Events from 1985 to 2017 (Brakenridge, 2010). All flood events not driven by surface runoff (Jökulhaup, tsunami, tides, avalanche, storm surge, barrier break or release, ice jam or ice break-up, or ice melt) were ignored. For a given flood event, point and areal locations representing flooded surfaces are accumulated through news, government, and instrumental sources. The area affected is then defined as the polygon that bounds all of these points and areas. For particularly damaging floods, the flooded areas, and so the resulting bounding polygon, are validated by satellite observations using the MODIS sensors, which operate at a resolution of 250 m. Although the bounding polygon inherently overestimates the true surface flooded, it represents the synoptic and climatic scales over which riverine flood is driven and so is an appropriate quantity with which to associate climate variables to observed large-scale floods.

Each flood is further characterized by impact metrics such as the duration, deaths caused, population displaced, and severity (a proxy of return period). We used the population displaced to construct our impact database and model. The reported population displaced in DFO depends on the nature of the source used. Generally, the value reported by the original news source is the number of people left homeless or

Table 2
Forcing Climate Variables From the NCAR CESM Large Ensemble (CESM-LE) Community Project

Variable long name	Variable short name	Model quantity
Atmospheric rain	RAIN	Precipitation
Atmospheric snow	SNOW	Precipitation
Reference height temperature	TREFHT	Temperature

evacuated. Moreover, when a categorical value such as “thousands” or “more than 10,000” is reported, DFO uses 3,000 and 11,000. DFO assumes that there are four people in a family for the cases when families displaced are reported.

Flood hazard and risk are defined over the HydroBASINS (Lehner & Grill, 2013) Pfafstetter level 5 watersheds, which unless otherwise stated we refer to simply as watersheds. The databases for flood occurrence and impact developed here are therefore defined over these 4734 watersheds that cover the entire global land surface except Antarctica (Figure 2). To

associate historical flood occurrence and impact to the driving environmental quantities, we take the spatial intersection of the DFO events and the HydroBASINS watersheds. This results in a new set of polygons which we refer to as watershed-floods. DFO flood events generally occur over several watersheds and therefore result in one or more watershed-floods.

Since the DFO event polygons are constructed by finding a boundary around locations of observed flood, they overestimate the true surface that was flooded. To avoid identifying watershed-flood events in watersheds where the intersection is minor, we ignore watershed-floods that make up less than 5% of the watershed area. We investigated a subset of DFO events with zero reported people displaced and treated such floods as missing data instead of events with zero impact. This resulted in 3,160 of the 4,499 DFO floods with at least one person displaced (Figure 3a). For each DFO flood event, we distributed the reported population displaced over the watershed-floods in proportion to the population of the watershed in the year of the flood (Figure 3b).

To associate flood occurrence and impact with time-varying environmental predictors, we used the following criteria: (1) global spatial coverage, (2) temporal coverage that contained the DFO flood event dataset (1985–2017), (3) horizontal spatial resolution less than 1°, (4) at least daily temporal resolution, (5) resolved in the CESM-LE climate product that drives the catalog simulation. While observational products such as soil moisture (Gruber et al., 2019) and terrestrial water storage (Tapley et al., 2004) were of interest, our criteria limited our analyses to precipitation and temperature products.

Precipitation is the key driving predictor of flood occurrence and impact. We represent it using the Climate Hazards Group Infrared Precipitation with Stations (CHIRPS) dataset (Funk et al., 2015) for latitudes ranging from 50°S to 50°N, and the CPC Global Unified Gauge-Based Analysis of Daily Precipitation (CPC Precipitation) dataset (Xie et al., 2007) for all other latitudes. Temperature plays a key role in evapotranspiration

Observed DFO Flood Events

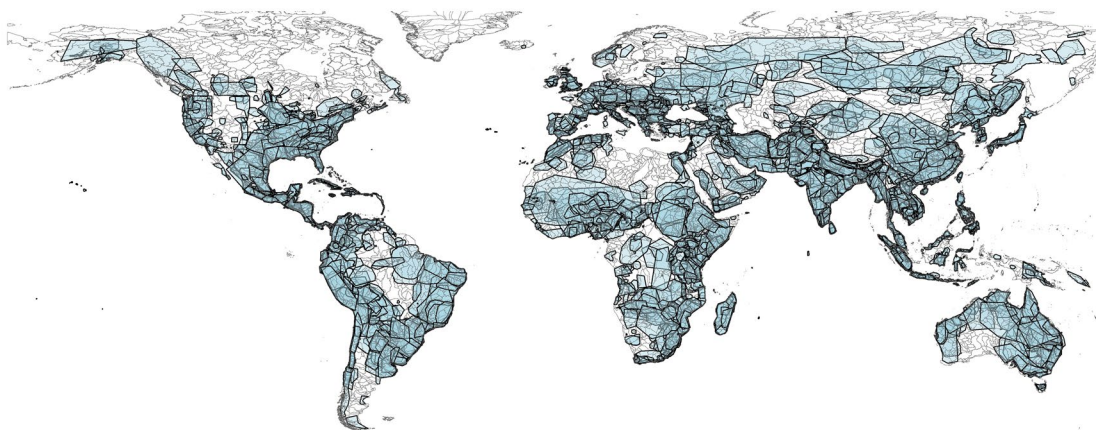


Figure 2. Observed Dartmouth Flood Observatory (DFO) flood events from 1985 to 2017. Transparent blue polygons represent the bounding area of the Dartmouth Flood Observatory (DFO) Global Active Archive of Large Flood Events floods. HydroSHEDS watersheds at Pfafstetter level 5 are white background polygons outlined in gray. The spatial intersection of these two sets of polygons results in the watershed-floods which we use to construct the flood occurrence and impact databases.

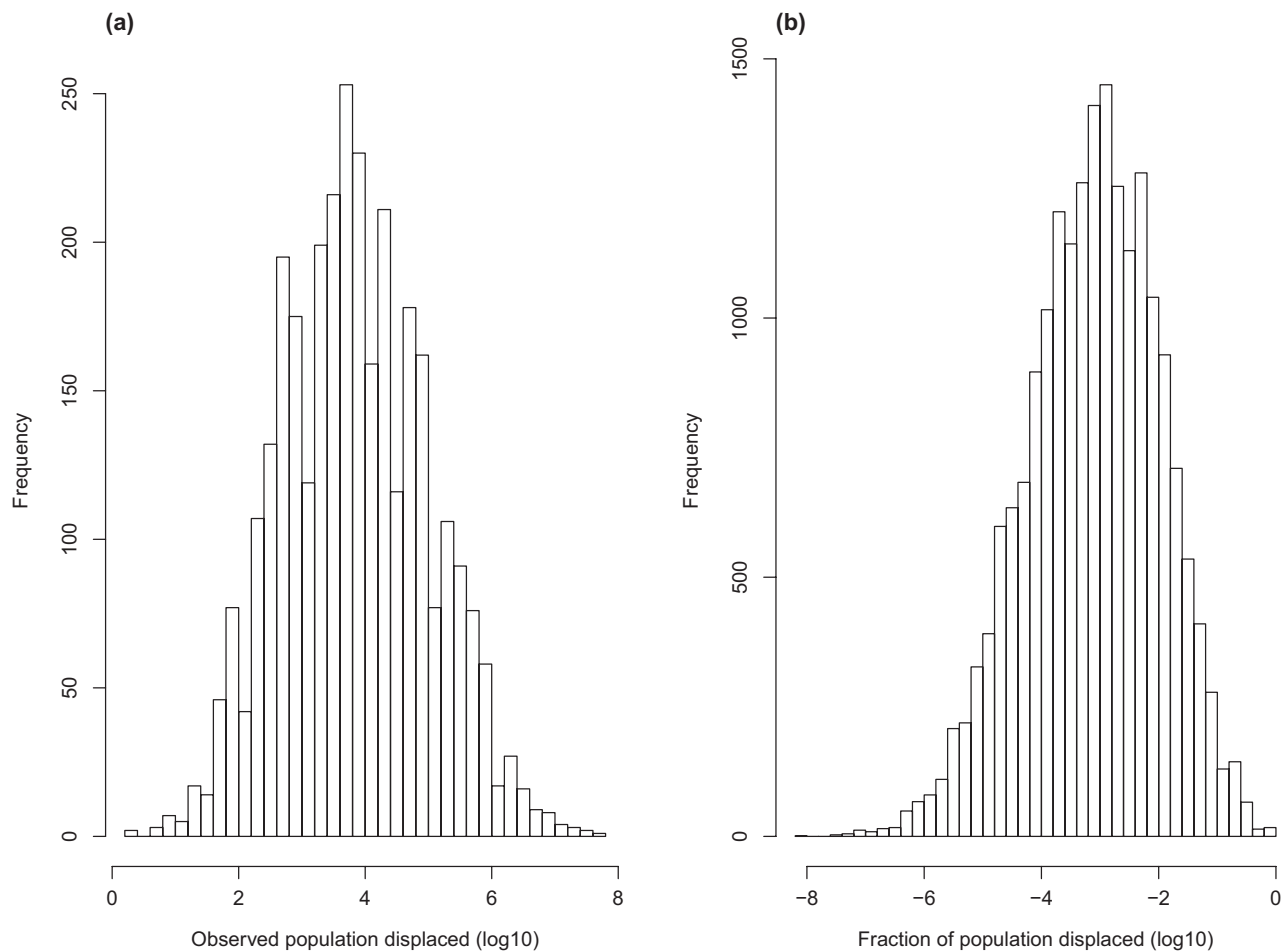


Figure 3. Distributions of population displaced impact variable. (a) Observed population displaced ($\log_{10}$) for each DFO event (3,160 non-zero observations). (b) Fraction of watershed population displaced ($\log_{10}$) for each watershed-flood event (19,746 observations).

(Li et al., 2016) and we represent it with the CPC Global Daily Temperature (CPC Temperature) dataset (Shi, 2007). The time-varying climate datasets we used are resolved daily; however, since the relevant temporal scales for the flood occurrence and impact models are different, we applied the daily temperature and precipitation predictors differently. To capture short, medium, and long-term temporal dependence that ranges from days to an entire season, we considered the following four timescales of temperature and precipitation by calculating the average temperature and precipitation over the previous 7, 8–30, 31–60, and 61–120 days. For the occurrence model, which models the annual probability of flood in a watershed, we consider annualized predictors. Temperature is represented by the watershed annual average, whereas the hydrological annual maximum precipitation is used for the four timescales just mentioned. The impact model climate predictors are assumed to be the value of the predictor timescale on the final day of the observed flood.

The remaining environmental predictors represent watershed storage capacity. Watershed topography is a crucial characteristic in hydraulic modeling that determines flows along the surface and subsurface and influences infiltration into the subsurface (Farr et al., 2007). We represent topographic features and watershed location with the HydroBASINS (Lehner & Grill, 2013) and High-resolution global topographic index values (TI) (Marthens et al., 2015b) datasets. Land cover characteristics alter infiltration into the subsurface and can additionally influence evapotranspiration (Nie et al., 2011). We represent land cover with the Global Land Cover SHARE (GLC-SHARE) (Latham et al., 2014) dataset. Soil permeability affects infiltration (Wolock et al., 2004) and subsurface drainage (Yu et al., 2000), whereas soil depth is indicative of soil water storage. We represent soil type with the Global Soil Database (GSD) (Shangquan et al., 2014) and depth with

the Global Depth to Bedrock (GDB) (Shangguan et al., 2017) dataset. Bedrock porosity and permeability proxy subsurface storage and drainage, respectively (Wolock et al., 2004). We represent these quantities using the GLObal HYdrogeology MaPS 2.0 (GLHYMPS 2.0) (Gleeson et al., 2014) dataset.

The exposure predictors population density and gross domestic product (GDP) per capita are interpreted as time-varying proxies of urbanization and flood control that can additionally capture features of land cover and socioeconomic change that are not represented in the time-invariant GLC-SHARE dataset, and as such affect both hazard and impact. Population is represented with the Gridded Population of the World (GPW) (Doxsey-Whitfield et al., 2015) (Figure S3) and Anthropogenic land-use estimates for the Holocene (HYDE3.2) (Klein Goldewijk et al., 2017) datasets, and wealth with the Gridded global datasets for GDP and HDI (Human Development Index) over 1990–2015 (GDPHDI) (Kummu et al., 2018). Temporal extrapolations of these variables are described in the supporting information. Table 1 details the datasets used in the occurrence and impact database development.

Our work was coded using the R software environment (R Core Team, 2018). We used the package *data.table* (Dowle & Srinivasan, 2018) for data processing and merging, and *velox* (Hunziker, 2017) to calculate aggregations of the predictor and climate variables over the HydroBASINS watersheds. The packages *sp* (Pebesma & Bivand, 2005) and *sf* (Pebesma, 2018) were applied for spatial analyses such as spatial intersections. *doSNOW* (Wallig et al., 2017) was used to parallelize computations. We used *cartography* (Giraud & Lambert, 2016) for choropleth maps and *RColorBrewer* (Neuwirth, 2014) for the color schemes.

3.2. Occurrence

Based on the observed watershed-floods, we assumed that a level 5 watershed can be in one of two states (“flood” or “no flood”) in a given hydrological year. Given that there are 4,734 watersheds and that we use 32 hydrological years (October 1–September 30) from the DFO database, there are up to 151,488 occurrence observations that can be either “flood” or “no flood.” We associated each observation to 38 climatic, hydrological, and socioeconomic predictors described in Section 3.1 and listed in Table 1. Removing observations with missing data in the predictors (mostly high latitude sites lacking topographical observations) left us with 128,494 observations for the fitting process.

The statistical problem for flood occurrence is a classification problem with the binary response variable (“flood”, “no flood”). We considered classical and machine learning methods such as logistic regressions (LR), random forests (RF, Breiman (2001)) and artificial neural networks (NN, McCulloch and Pitts (1943)) to solve this problem. Hastie et al. (2009) discuss the three methods in chapters 4, 15, and 11, respectively. Since it is difficult to define explicit functional forms and interaction terms between the predictor variables a priori, in particular given the variety of flood regimes that exist globally, we adopted a machine learning approach that builds such relationships from observations using a random forest. Validation of the occurrence models are presented in Section 5.1 for the random forest, and in the supporting information for the other models.

We first fit each of the three models (LR, RF, and NN) globally, thereby providing one set of parameter estimates per model. To guarantee that local dynamics were appropriately represented in the global fit, we also fit each model over aggregations of watersheds at Pfafstetter level 2 (HS2). Since there are 62 watersheds at HS2, 62 parameter sets were fitted. For the global occurrence fit, we took a random sample of 70% of the observations as the training set and used it for all models. We then conducted an out-of-sample validation on the remaining 30% of observations, the test set (Figure S1). For the RF model, we considered 500 decision trees and used five randomly selected predictors, the square root of the 38 predictors, for each split. There are popular rules of thumb for the choice of this parameter such as the square root or one third of the number of predictors. However, this is a hyperparameter that should be tuned in practice (Hastie et al., 2009). We began with the square root of the number of predictors and in testing alternative values, we found there to be negligible differences in the model results and so maintained this choice. To compare the regional and global fits, we used the same training and test sets for the regional fits by HS2. In applying empirical fits in each HS2, we split our observations into 62 sub-samples. Given the resulting smaller samples, for some regions there are insufficient observations to conduct a fit. An extreme example of such a case would be that there are no flood observations in a region. To guarantee a sufficient sample of flood events, we only conduct

a fit if more than 3% of observations in the region are “flood.” The structure of the regional RF models follows the global model, with 500 decision trees and 5 predictors considered for each split. Methodological details for the NN model are provided in the supporting information.

A good classification model should predict an event when there is really an event (a true positive). However, when occurrences are rare, it is easy to be accurate most of the time by simply predicting the event always (or never) occurs. As such, one needs to evaluate models by balancing true positives (TP) and true negatives (TN) with false positives (FP) and false negatives (FN) (Fawcett, 2006; Powers, 2011). Such analyses are commonly summarized using the receiver operating characteristic (ROC) curve, a plot of the true positive rate ($TP/(TP + FN)$) versus the false positive rate ($FP/(FP + TN)$), which are both determined as functions of the cutoff probability used to define a predicted “flood.” The area under the ROC curve (AUC) is a summary measure that ranges from 0 to 1 and indicates the likelihood that the classification model can differentiate between “flood” and “no flood.” An AUC above 0.5 indicates that the model has the ability to differentiate between classes, yet a value above 0.7 is often used to identify a skilled model (Buizza et al., 1999). We report the AUC aggregated over HS2 watersheds if there are at least 10 observations and at least five floods in the test set (Section 5.1). Other model evaluation metrics are discussed in the supporting information. The ROCR package (Sing et al., 2005) was used for the occurrence model validation calculations.

For the RF models, we consider two variable importance measures. The first considers how the accuracy (fraction of observations that are correctly classified $(TP + TN)/(TP + TN + FP + FN)$) changes in reaction to separately permuting the observations of each predictor in the out-of-bag (a test set) observations. The second measure employs node purity in terms of the Gini impurity index (Hastie et al. (2009)), which rewards models with homogeneous predictions.

3.3. Impact

For each watershed-flood, flood impact was represented as the fraction of the watershed population displaced (displaced population in watershed-flood/watershed population). Since the fraction of population displaced spans eight orders of magnitude, we applied a log10 transformation (Figure 3b). Overall, the impact model aims to explain the log10 of the fraction of population displaced as a function of demographic, wealth, climatological, and watershed predictors. The 41 predictor variables used to model this fraction are the same as for the occurrence model, with the important difference that the driving temperature and precipitation predictors are taken to be those for the final day of the event as reported in DFO (Table 1). The climatological and watershed predictors aim to capture the effects of the intensity of a flood on the fraction of population displaced whereas population density and wealth per capita aim to capture the vulnerability of a population. There are 19,746 complete observations in the impact database with at least one person displaced and no missing data in the predictors. While we used population displaced to represent flood impact, our approach is flexible enough that it can be used to build models with alternative impact quantities such as economic loss or disruption so long as the impact observations can be matched to DFO events.

As in the statistical modeling for occurrence, we considered classical and machine learning methods, namely linear regressions (LR), random forests (RF) and neural networks (NN) (see above references for details). The global impact model fitting procedure follows the occurrence model and takes 70% of the observations to form the training set. All validations are done by applying the fitted model to the remaining 30% of observations (Figure S2). For the RF fits, since there are 41 predictors, we consider six randomly selected predictors at each tree split and repeat this for 500 regression trees. As with occurrence, we conducted regional fits of the impact observations. A regional model was fitted when there were at least 30 watershed-flood impact observations in the HS2 watershed. Validation of the impact models are presented in Section 5.2 for the RF model, and in the supporting information for the other models.

To assess the quality of the impact models, we considered two metrics using out-of-sample observations: the root mean square error (RMSE, lower is better) and the R-squared (higher is better, with 1 being the maximum). The RMSE summarizes the model error whereas the R-squared measures the proportion of variance explained by a model (Hastie et al., 2009) (Section 5.2). For the RF model of flood impact, we additionally measure variable importance. The mean squared error is calculated for permutations of each predictor in the test set is used to represent accuracy, whereas the residual sum of squares is used to measure node purity.

The linear model and logistic regression fits were achieved with the stats package core functions `lm()` and `glm()` (R Core Team, 2018), and we applied the randomForest (Liaw & Wiener, 2002) and RSNNS (Bergmeir & Benítez, 2012) packages for the RF and NN models. The randomForest package was additionally used for the variable selection methods for the RF models.

4. Flood Event Simulation

4.1. Climate Model Output Data

The NCAR-CESM1 (Hurrell et al., 2013) Large Ensemble (CESM-LE) (Kay et al., 2015) is a set of 40 simulated trajectories of the Earth system for the years 1920–2100. This Earth system model represents the dynamics and interactions between the atmosphere, ocean, land surface, and cryosphere, and outputs a variety of variables for each component. We apply the precipitation (rainfall + snowfall) from the Community Land Model 2.0 (Lawrence et al., 2011), as well as the temperature 2 m from the surface from the Community Atmosphere Model 5.2 (Neale et al., 2012) (Table 2).

We forced the empirical occurrence and impact models with output from this Earth system model to simulate a large catalog of flood events. The CESM-LE was designed to examine interannual climate variability in the context of anthropogenic climate change. Each member is initialized with a roundoff error perturbation to the atmosphere in model year 1850, so that by 1920 the members are independent of one another yet driven by the same forcing of radiative gases in the atmosphere and aerosols. The 40 members therefore represent 40 potential realizations of the Earth system from 1920 to 2100. To be consistent with the flood observational record from DFO, we used only the years 1980–2020. This results in 40 years from each of 40 members, which we refer to as 1,600 member-years of global climate model output. The NCAR-LE employs a single historical forcing scenario up to 2005 and the RCP8.5 climate scenario from 2006 to 2100. Evidence is mounting that the RCP8.5 climate scenario projection used in the CESM Large Ensemble is highly unlikely looking to the future beyond 2030 (Hausfather & Peters, 2020). Nevertheless, the RCP8.5 scenario closely followed observed accumulated CO₂ (Schwalm et al., 2020) from 2006 to 2020. Given that we focused our global flood catalog on the years 1980–2020, we feel that the use of the CESM Large Ensemble is appropriate.

As with the predictor variables used in the statistical fit step, for each climate forcing variable (Table 2) we aggregated by averaging over the grid points in each level 5 watershed. CESM-LE atmospheric rain and snow are summed to model precipitation. Snow is assumed to melt immediately to water when it touches the ground at a bulk weight density of 100 kg m⁻³ corresponding to that of fresh snow (Meløysund et al., 2007). Although we do not explicitly represent snow accumulation, by tracking the accumulation of the water equivalent of snowfall by means of the longer timescales mentioned above (31–60 and 61–120 previous days of precipitation), in addition to temperature over these timescales, we implicitly captured the seasonal snow accumulation signal that drives snowmelt floods.

Biases in the CESM-LE precipitation are corrected relative to CHIRPS and CPC Precipitation, and biases in temperature are corrected relative to CPC Temperature. We apply the methodology of Hempel et al. (2013) to precipitation and temperature aggregated over each level 5 watershed. This bias correction approach is widely used in the hydrological and flood impact literature and corrects monthly means and daily variability about the means, thereby preserving long-term observed trends. First, monthly mean temperature and precipitation are corrected using additive and multiplicative factors, respectively, that are constant over time and so preserve observed trends in the historical record. Daily variability in the temperature anomaly (departure from monthly mean) is corrected using a linear regression on ranks. To resolve the problem of drizzle in climate model precipitation (far fewer simulated zero-precipitation days than observed), the frequency of dry months and dry days within wet months is corrected. When correcting the number of dry months, we instead use the fraction of observed dry months since the observational dataset (CHIRPS, with 37 years of data) is smaller than the simulated one (CESM-LE, with 40 years of data). Finally, daily variability in normalized monthly precipitation is corrected using a non-linear regression on ranks.

Table 3
Duration of Simulation Computation Steps

Step	Description	Duration	Repetitions
1	Occurrence - Simulation	1.34 s/simulation	625
2	Intensity - Distribution mean	36.9 min/member	40
3	Intensity - Distribution standard deviation	3.1 s	1
4	Intensity - Merge mean and standard deviation	6.8 s	1
5	Simulate intensity and combine with occurrence	3.75 s/simulation	50
6	Calculate standard deviation correction	1.68 min	1
7	Repeat step 4 with corrected standard deviation	6.8 s	1
8	Simulate corrected intensity and combine with occurrence	3.75 s/simulation	625

4.2. Occurrence

To simulate floods we apply the fitted occurrence model to compute predicted flood occurrence probabilities (1,600 member-years $\times$ 4,734 watersheds) using the bias corrected precipitation and temperature outputs from the CESM-LE and the watershed-scale environmental and socioeconomic variables. Flood occurrence, for each member-year and each watershed, thus follows a Bernoulli distribution whose only parameter represents the probability of a riverine flood. The flood occurrence probability is unique for every ensemble member year and watershed (40 members $\times$ 40 years $\times$ 4,734 watersheds). To generate a large sample of flood events at the global scale, we sample from each of these Bernoulli distributions 625 times, resulting in a million simulated years.

4.3. Impact

For each simulated flood event, we in turn simulate impact as the fraction of the watershed population displaced. There is no inherent randomness in the random forest (RF) impact model, and so the predicted fraction of population displaced will always be the same for the same set of covariates. To integrate randomness, we assume that the log10 of fraction of population displaced is normally distributed (Figure 3b). For each watershed, we take the mean parameter of the flood impact distribution to be the maximum of the daily impact predictions of the RF impact model (Section 3.3) over the hydrological year. The standard deviation parameter is calculated as the root mean squared error of the impact model predictions relative to the observations. We observed increasing and then constant prediction errors for watershed-floods as a function of the population density over the watershed (Figure S10), and so separated the sigma values into one of 9 equally-sized groups as a function of watershed population density. Given the mean and standard deviation of the normal distribution for each watershed and ensemble member year, we simulate an impact value for each simulated flood over the million years of the catalog.

Table 3 details the steps involved in simulating the flood catalog. The durations reported are for a single processor thread on an Intel Xeon E5-2650 v3 at 2.30 GHz. Step 2 takes $\sim$ 1 day to compute daily impact predictions over the model years 1980–2020 of the CESM-LE. While this is the most computationally demanding step, it takes only 37 min per ensemble member and so the duration of the user's calculations depend on the number of ensemble members of interest. All other steps are rapid, with the simulation of 1 million years of flood occurrence requiring only 14 min (Step 1) and the simulation of corrected impacts and merging with occurrence taking only 39 min (Step 8). While we worked with a single thread, step 2 can be parallelized given sufficient system memory, easily reducing the calculation by a factor of 6–8.

5. Model Evaluation

We describe the results of the occurrence and impact models in terms of goodness of fit, parsimony, ease of use, and interpretation in Sections 5.1 and 5.2. Results are presented on world maps of aggregated HydroBASINS watersheds. We analyze the realism of the entire simulated flood catalog in Section 5.3. Additional validations and other results can be found in the supporting information.

Table 4
Out-Of-Sample Area Under the ROC Curve (AUC) for Occurrence Models and Fitting Approaches Considered

Model	Global Fit	HS2 Fit
LR	0.735	0.789
RF	0.788	0.792
NN-24	0.787	-
NN-12-12	0.781	-
NN-24-24	0.776	-
NN-12	-	0.777

5.1. Occurrence

For global fits, all three models show useful predictive skill since the AUC are larger than 0.7 (Table 4). However, random forest (RF) and neural network (NN) models have consistently better predictive skill than logistic regressions (LR), with AUC of 0.78 as opposed to 0.73. Superior precision and recall for the RF and NN global fits further indicates the predictive advantage of working with these models (Figure S4). Since the RF and NN models are built to capture complex non-linear relationships and interactions between predictors and, given the problem at hand, it is not a surprise that such non-linearities and interactions appear. As such, RF and NN perform similarly globally and over regional aggregations of watersheds. The quality of logistic regressions improves when fitted regionally in terms of the AUC (Table 4, Figure S5). That said, for fits over HS2 watersheds, all three methods perform similarly.

Neural networks do not significantly improve the quality of the fit when compared to RF models. With only 128,494 observations, the dataset is likely not in the appropriate sample size regime to observe the benefits of NNs. Given the difficulty in interpreting NNs, we cannot recommend their use for this application. We generally find that the RF, fitted globally or regionally, is a solid approach in terms of in-sample and out-of-sample fit of global flood occurrences. Given that the RF is composed of individual decision trees applied to random samples of the observations, it is also easier to interpret than the other models. Despite important differences in methodology, we are encouraged that Li et al. (2019) also find that the RF model has the most skill for flood occurrence predictions. If one seeks (statistical) confidence intervals on model parameters, we recommend the use of logistic regressions in combination with regional fits. Figures S6 and S7 present the AUC for each HS2 watershed for the LR and NN models, respectively.

Figure 4 demonstrates the performance of the globally fitted RF in terms of the area under the ROC curve aggregated over HS2 watersheds. The occurrence model does particularly well in Eastern Canada, Eastern China, Japan, Indonesia, Australia, and the Amazon. While we cannot calculate an AUC for 16 of the 62 HS2 watersheds due to insufficient observations, the global RF model possesses useful predictive skill (AUC more than 0.7) in 28 of the remaining 46 HS2 watersheds and so generally has the ability to identify the environmental and socioeconomic features that generate flood occurrence. We are encouraged that none of the HS2 watersheds have an AUC less than 0.5, and that there are no systematic spatial biases exhibited.

AUC RF

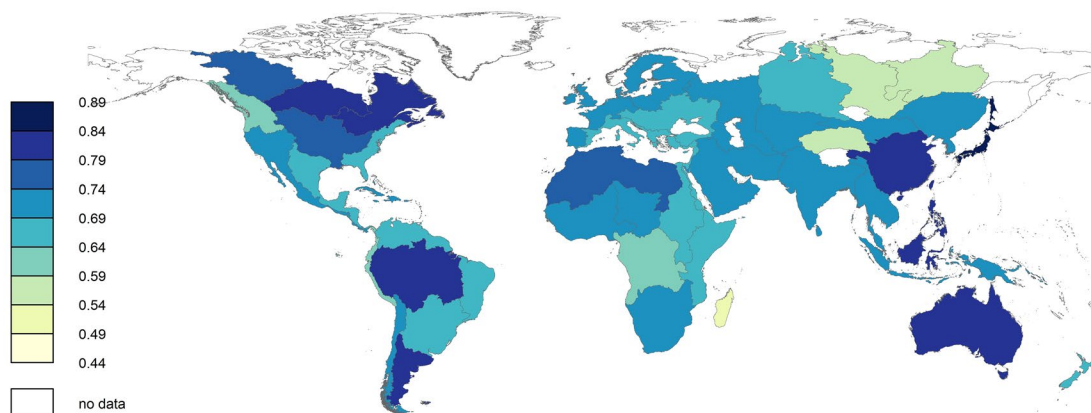


Figure 4. World map of the area under the receiver operating characteristic curve (AUC) for the globally-fitted Random Forest (RF) model. The AUC is evaluated using out-of-sample observations aggregated over each level 2 watershed (HS2) and indicates the probability that a model can differentiate between “flood” and “no flood.”

Table 5
Most Significant Predictors of Flood Occurrence Ranked by Two Methods for the Globally Fitted Random Forest (RF) Model

	Mean decrease in accuracy	Mean decrease in Gini impurity index
1	GDP per capita	Population density
2	Population density	Precipitation (previous 7 days)
3	Temperature (annual mean)	GDP per capita
4	Precipitation (previous 7 days)	Precipitation (previous 8–30 days)
5	Topographic position index	Precipitation (previous 31–60 days)
6	Precipitation (previous 8–30 days)	Precipitation (previous 61–120 days)
7	Gravel	Temperature (annual mean)
8	Precipitation (previous 61–120 days)	Cropland
9	Aspect	Latitude
10	Porosity	Longitude

Table 5 lists the 10 most significant predictors found to explain flood occurrence with the globally fitted RF model. Regardless of the measure chosen, the two exposure predictors of GDP per capita and population density are prominent. Unsurprisingly, the precipitation over various timescales are key predictors. Annual mean temperature is also an important predictor, driven by its link to interannual patterns in evapotranspiration. The remaining predictors represent the residual components of flood hydrology. Topographic effects are represented by the topographic position index and aspect, soil content by gravel, and land usage by cropland. Bedrock porosity provides a proxy of longer term storage.

5.2. Impact

As with the occurrence model, we fit each model globally and by HS2 regional subsets of watersheds. In terms of the quality of the out-of-sample fit, we again cannot claim that the NN models outperform the others, so given their complexity we do not recommend them. Linear models (LM) perform poorly globally but their fit improve once fitted over regional subsets of watersheds (Table 6). However, LM are outperformed by RF in terms of global and local fit and hence we recommend the latter method for this application. The supporting information presents the RMSE and R^2 for all three models fit globally (Figure S8) and by regional subsets of watersheds (Figure S9).

Figure 5 shows the root-mean-square error (RMSE) aggregated by HS2 watershed regions for the RF fitted globally. Almost all HS2 regions have an out-of-sample RMSE below 1, which given the log10 scale of the fraction of population displaced used as the response variable, indicates that the predictions are within an order of magnitude of the observations. This result is excellent given that impact ranges over eight orders of magnitude (Figure 3b). It is also important to note that in the observations, the fraction of displaced people is very small (mainly in the range of $[-5, -2]$ on a log10 scale) meaning that absolute errors, rather than

relative errors, are quite small. We validated the global impact fit over HS2 watersheds (Figures 6 and S8) for 39 of the 62 HS2 watersheds where there were sufficient observations (more than 10) in the test set. We found that 10 of the HS2 watersheds analyzed have an out-of-sample R^2 below 10%, while 15 are above 20% and five above 40%. Given the challenge of predicting the fraction of the population displaced by a flood over the globe, we find these to be promising results. A number of the poorer predictive results are influenced by limited observations, such as for Madagascar, Papua New Guinea, and New Zealand (see Figure S2). However, for the more challenging watersheds the predictors are not representing the fraction displaced because: (1) Other predictors represent the relevant flood regimes in those watersheds, (2) there are biases or inaccuracies in the impact observations, (3) there are biases in the manner in which the

Table 6
Out-Of-Sample R^2 for Impact Models and Fitting Approaches Considered

Model	Global Fit	HS2 Fit
LM	0.172	0.209
RF	0.325	0.353
NN-26	0.291	-
NN-13-13	0.287	-
NN-26-26	0.303	-
NN-13	-	0.308

RMSE RF

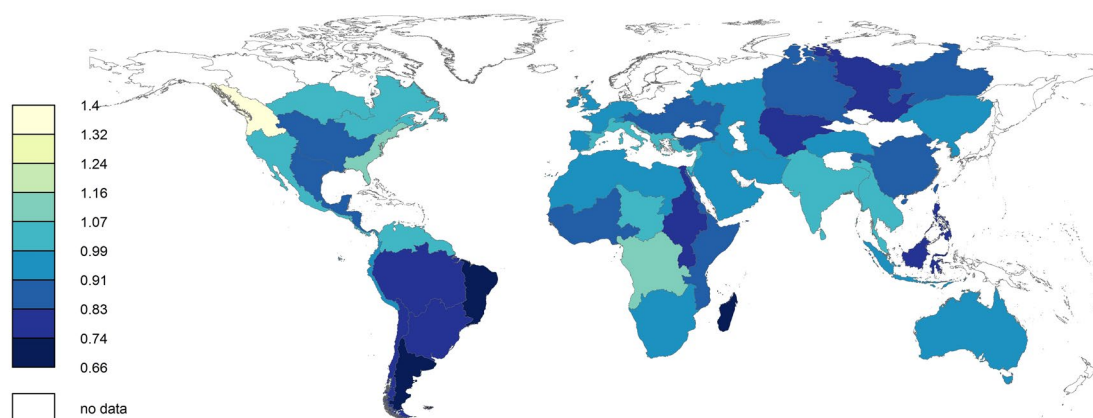


Figure 5. World map of the out-of-sample RMSE for each level 2 watershed for the flood impact model fitted with a random forest (RF).

persons displaced are associated to particular watersheds. For the impact model, we are reassured by good levels of out-of-sample variance explained over the majority of the globe, and by a lack of systemic patterns in the watersheds with poorer variance explained.

Table 7 lists the 10 most significant predictors in the globally fitted RF impact model. Regardless of the measure used, we find that most timescales of precipitation and temperature variables need to be included in the model, as well as population density and GDP per capita. The resulting predictors are similar to those identified in the occurrence model, with lagged temperature predictors taking on an important explicative role. Overall, we find that lagged precipitation and temperature variables, when applied with the exposure predictors of GDP per capita and population density, capture the majority of the resolved signal of seasonal flooding.

5.3. Global Model

This section assesses the entire flood catalog, which consists of 1 million simulated years of flood occurrences and impacts over 4,734 watersheds globally. We evaluate the simulated occurrence and population displaced. Despite the bias correction of precipitation and temperature, the time-varying CESM-LE output and watershed-scale predictors do not sufficiently capture impact extremes. This results in underestimated densities for both low and high values of the fraction displaced (Figures S11 and S12). To remedy this issue

Table 7
Most Significant Predictors of Impact Model Ranked by Two Methods

	Increase in mean squared error	Increase in residual sum of squares
1	Precipitation (previous 8–30 days)	GDP per capita
2	GDP per capita	Temperature (previous 61–120 days)
3	Precipitation (previous 7 days)	Temperature (previous 31–60 days)
4	Population density	Temperature (previous 8–30 days)
5	Precipitation (previous 31–60 days)	Temperature (previous 7 days)
6	Precipitation (previous 61–120 days)	Population density
7	Temperature (previous 61–120 days)	Precipitation (previous 7 days)
8	Temperature (previous 7 days)	Latitude
9	Longitude	Precipitation (previous 31–60 days)
10	Temperature (previous 31–60 days)	Precipitation (previous 8–30 days)

in a simple way, we scaled the standard deviation parameters of the impact distribution to match the observed standard deviation (Figures S11 and S12).

Figure 7 compares the mean simulated and observed occurrence and impact aggregated over level 3 watersheds (HS3). It is important to note that such a comparison is ambitious since the performance of the global model depends on the performance of the CESM-LE to generate conditions favorable for flood. As such, the clustering of points along the 45° lines indicates that the global model works well overall to simulate a realistic number of floods and displaced people compared to the observations. For low risk areas though, the model tends to slightly overestimate flood hazard. This could be because no flood was observed over the 32-years observational record even if the true flood probability is non-zero. This bias results in overestimates of the number of displaced people for small population watersheds. Nevertheless, for HS3 watersheds with an annual mean of more than 300 people displaced (the great majority of watersheds where risk analysis is relevant), the model predictions fit the observations well.

The world map in Figure 8 illustrates the average error in the number of displaced people. The darkest blue shade represents locations where the overestimation is larger than a factor of 10. Of these regions, we observe that they are either in far northern regions (Northern Canada, Sweden, Finland, North-Western Russia) or dry climates (Libya, Egypt, Saudi Arabia, and Southwest Australia). While flood is not common in these regions, biases could be due to underreporting in DFO in these regions as they are sparsely populated, to overestimates in the CESM-LE flood-generating conditions, or a lack of fit of the occurrence/impact statistical models due to a lack of observations. Looking back to Figures 4 and 6, the fit of both statistical models is good in dry climates but not as successful over northern regions. Overall, there are less cases of underestimated population displaced. The model underestimates population displaced in a few South Asian watersheds in Bangladesh, India, and Nepal. This region, particularly Bangladesh, is known for extreme floods resulting in millions of displaced driven by the unique combination of precipitation extremes from the annual Indian Monsoon, low-lying and complex hydrology, high population, and poor infrastructure (Dewan, 2015), which is beyond the ability of our model to capture.

6. Global Flood Hazard and Risk Maps

To illustrate potential uses for the model, we show global flood hazard and risk maps from the stochastic catalog comprised of 1 million years of events. Figure 9 presents the annual mean flood frequency in the catalog for each HS5 watershed, which can be interpreted as the annual flood probability over each HS5 watershed. Darker colors point to flood hot spots, with the highest flood probabilities in northeastern India, Bangladesh, and Myanmar being driven by the annual South Asian monsoons. Southeastern China, Japan, and certain areas of Southeast Asia are also high hazard areas. Other regional peaks are found in eastern USA, southern Mexico and Central America, southern Brazil, and parts of Europe. Although these probabilities seem high, they represent the likelihood that at least one river within an HS5 watershed overflows sufficiently so that at least one person is displaced. It should not be compared to typical return periods.

The annual average population displaced over each HS5 watershed (Figure 10) combines flood hazard (flood probability) and exposure (population) to yield the average number of displaced people in any given year (flood risk). For example, over Northern India, Bangladesh and China, many highly densely populated watersheds yield an annual average number of displaced people of over 100,000. Over Central Europe, for example, many watersheds have an average annual number of displaced people of about 1,000. We expect this model of flood risk to be highly relevant to risk management and socioeconomic studies.

7. Summary and Conclusions

By integrating the Dartmouth Flood Observatory database of historical flood events over the years 1985–2017 with the HydroBASINS database of watersheds, we created novel flood occurrence and impact databases that describe flood frequency and intensity over a watershed in terms of the climatic, watershed, and socioeconomic drivers. We then fitted classical regression and machine learning techniques to these data, and adopted the random forest (RF) model fitted to observations at the global scale. Finally, we generated a

R2 RF

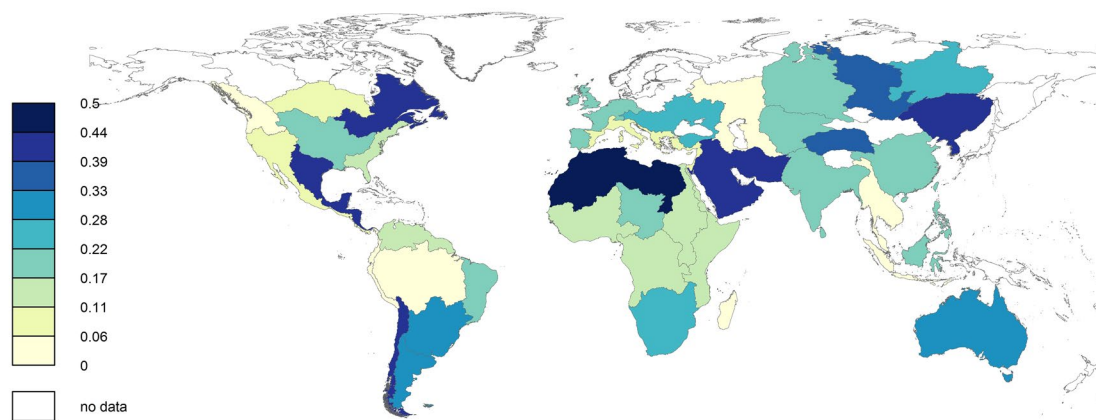


Figure 6. World map of the out-of-sample R-squared for each level 2 watershed for the flood impact model fitted with a random forest (RF).

global catalog of flood events by forcing the empirical model with bias-corrected precipitation and temperature output from the large ensemble of the NCAR CESM climate model.

The unique value of this global flood model lies in its ability to quickly simulate realistic flood events at a resolution (level 5 watersheds) that is useful for large-scale socioeconomic and financial planning. Trans-

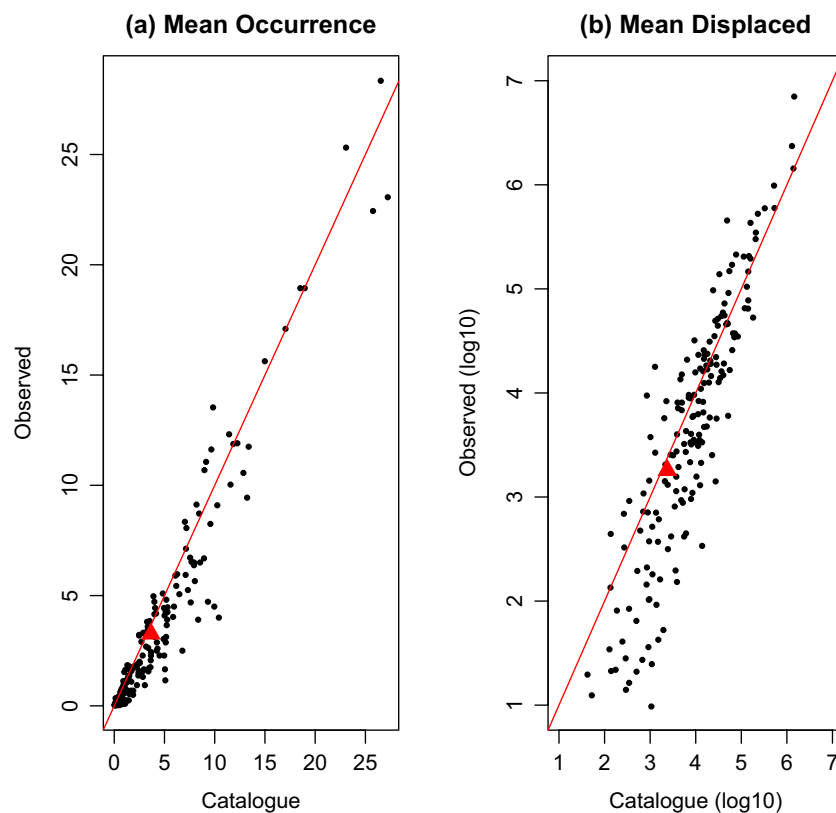


Figure 7. Observed versus simulated flood occurrence and population displaced aggregated by level 3 watersheds. (a) Mean observed flood frequency versus mean simulated frequency. (b) Mean observed versus simulated population displaced, expressed on a log10 scale. The red triangle represents the mean. Averages of observations are taken over the 32 hydrological years available from the DFO and over 1 million years for the catalogue.

HS3 log10 displaced error

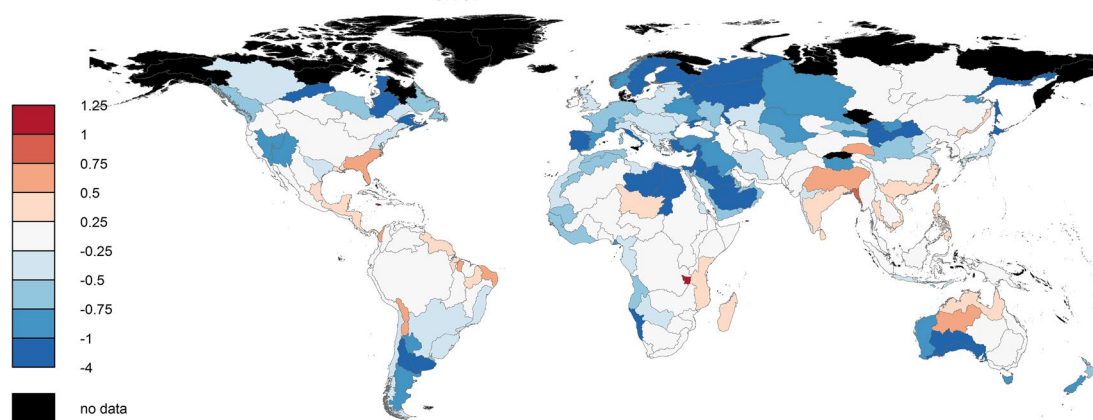


Figure 8. World map of the error on the mean number of displaced people. Represented as the mean observed displaced population less the mean displaced population in the catalog (with each member of the difference being on a log10 scale). Negative values indicate the model overestimates observations.

lating outputs from a climate model into flood events facilitates the creation of scenarios and projections of impacts over various time horizons. One could apply global weather or seasonal forecasts to simulate flood impacts over time horizons from days to months, or focus on different time horizons from the NCAR CESM climate model and investigate the impacts of climate change on flood hazard and risk. Alternatively, subsets of the catalog could be extracted to investigate the impacts of various climate oscillations on flood hazard and various measures of risk. Including population and wealth per capita in the model allows for sensitivity testing and experimentation of the dependence of flood hazard and risk to changes in the spatial population and wealth patterns. Finally, one could conveniently consider shocks to the outputs of the climate model (such as significant changes in precipitation over a given area in a year) and evaluate different climate scenarios, as will be required by regulators of the financial services industry. We expect this model to be a useful empirically-based though climatically-consistent complement to the mechanistic and other approaches available.

Flood hazard

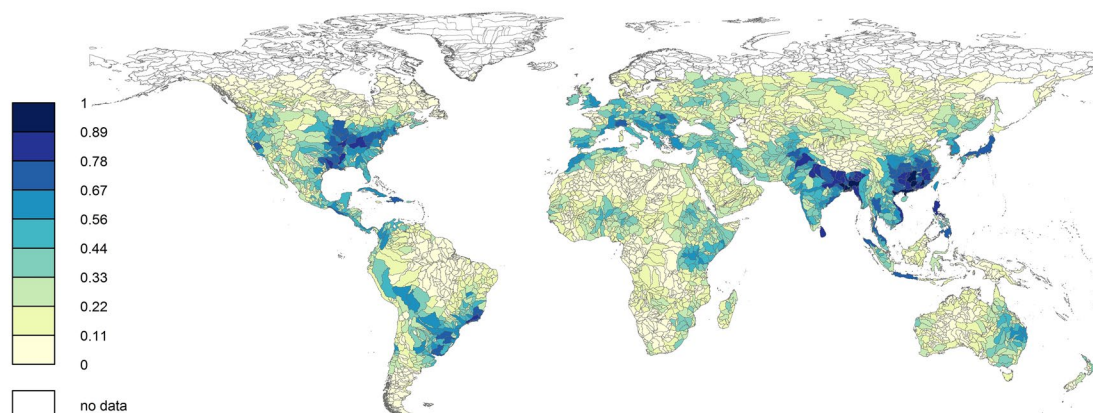


Figure 9. Global flood hazard map by level 5 watershed expressed as the annual flood probability.

Flood risk - Population displaced (log10)

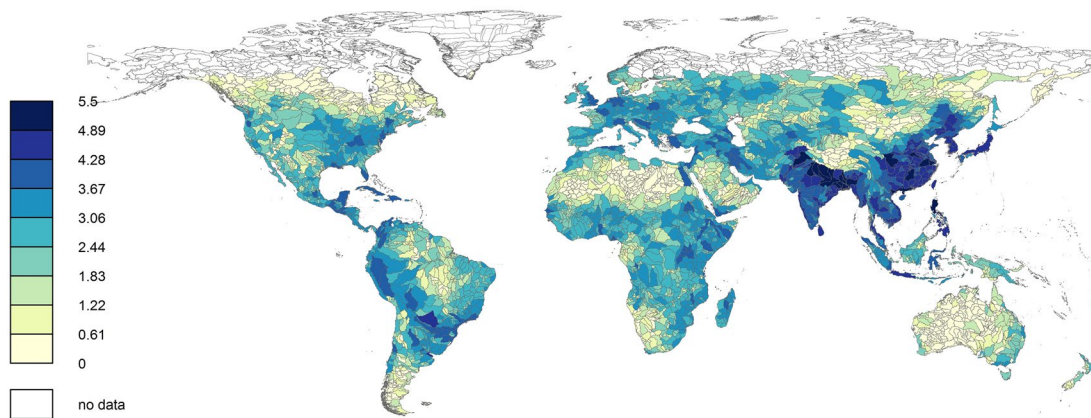


Figure 10. Global flood risk map, expressed as the average population displaced (log10) over the years 1980–2020.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study

Data Availability Statement

The HydroBASINS dataset is available at <https://www.hydrosheds.org/downloads>. The Dartmouth Flood Observatory (DFO) Global Active Archive of Large Flood Events can be downloaded at <https://floodobservatory.colorado.edu/Archives/index.html>. The CHIRPS global precipitation dataset can be downloaded at <https://data.chc.ucsb.edu/products/CHIRPS-2.0/>. CPC Global Temperature data provided by the NOAA/OAR/ESRL PSD, Boulder, Colorado, USA, from their Web site at <https://www.esrl.noaa.gov/psd/data/gridded/data.cpc.globaltemp.html>. CPC Global Unified Precipitation data provided by the NOAA/OAR/ESRL PSD, Boulder, Colorado, USA, from their Web site at <https://psl.noaa.gov/data/gridded/data.cpc.globalprecip.html>. The High-resolution global topographic index (TI) dataset is available at <https://doi.org/10.5285/6b0c4358-2bf3-4924-aa8f-793d468b92be>. The Global Soil Dataset for Earth System Modeling (GSD) can be downloaded at <http://globalchange.bnu.edu.cn/research/soilw>. The Global Depth to Bedrock Dataset for Earth System Modeling is available at <http://globalchange.bnu.edu.cn/research/dtb.jsp>. The GLOBAL Hydrogeology MaPS data is available at <https://doi.org/10.5683/SP2/TTJNU>. The FAO Global Land Cover (GLC-SHARE) Beta-Release 1.0 database available at <http://www.fao.org/land-water/land/land-governance/land-resources-planning-toolbox/category/details/en/c/1036355/>. The Gridded Population of the World (GPW) Version 4 Release 11 population count data is available at <https://sedac.ciesin.columbia.edu/data/set/gpw-v4-population-count-rev11>. The Anthropogenic land-use estimates for the Holocene; HYDE 3.2 dataset is available at <https://doi.org/10.17026/dans-25g-gez3>. Gridded global datasets for Gross Domestic Product and Human Development Index (GDPHDI) over 1990–2015 are available at <https://doi.org/10.5061/dryad.dk1j0>. The CESM Large Ensemble dataset is available at <https://www.cesm.ucar.edu/projects/community-projects/LENS/data-sets.html> and the authors acknowledge CESM Large Ensemble Community Project and supercomputing resources provided by NSF/CISL/Yellowstone. The data supplement to this article is available at <https://doi.org/10.5281/zenodo.3873422> and contains datasets, fitted statistical models, and an analysis script (Carozza & Boudreault, 2020).

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